Week 8

Objectives	We have three main objectives this week: Learn how to change the basis of a vector space. Understand the scalar product of vectors and master the Gram-Schmidt orthogonalization. **Solve the second and last questionnaire of Challenge 3**.
Session 22	As an introduction, watch the second part of the video "Vector Spaces". We start with an interesting example of a change of basis using the RGB and CMY color representations. Next, with another example, you'll see how to find the change-of-basis matrix that automatizes the changes of basis. Let's continue with the PDF resources. Continue reading sections 4.6 and 4.7 where the change-of-basis matrix of a vector space is described. Read carefully exemple 47, and check by yourselves all the steps in the resolution. Do self-assessment exercises 2, 5 and 17. Study the example on the change of basis from the UOC CalcME Guide.
Session 23	We will finish with Module 2 reading the scalar product of vectors (section 6). Read the definition, their properties and how to use it in order to analyze the relative positions of any pair of vectors. Finally, read the concept of orthonormal vectors, their properties and the Gram-Schmidt orthogonalization method. Do example 62 by yourselves.
Session 24	**This session doesn't have any specific program.** You can use this session for **solving the second questionnaire for Challenge 3**. With this questionnaire solved, we can close the third Challenge. Be sure to completely understand all the concepts we have worked here, since they will be useful for the next Challenge. Try to summarize your knowledge if necessary before ending this week. This is another good session to try practicing with a video explaining an exercise, for when you'll have to do it in the synthesis CAT.
Keep in mind	Be careful! This week you don't have as much work as the last one, but it is important for you to not relax too much. Once again, try to solve the exercises without looking at the solutions. Don't be afraid of "wasting" your time trying to solve them: any time you spend there will be very useful for your learning process. Remember that you should **solve the second questionnaire of Challenge 3** this week.